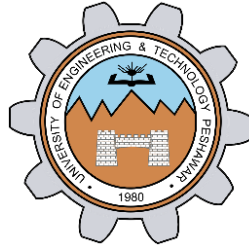


Interfacing PHP with MySQL

LAB # 06



Spring 2024

CSE-403L Database Management Systems Lab

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Engr. Sumayyea Salahuddin

Date:

17th May 2024

Department of Computer Systems Engineering
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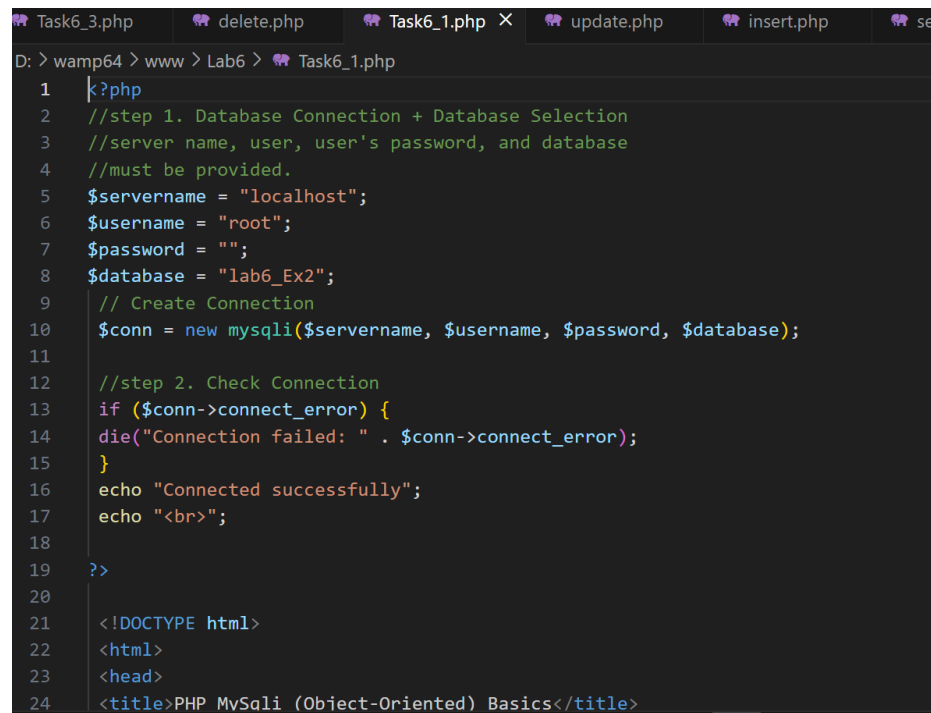
Objectives:

This lab aims at the understanding of:

- Type of MySQL Drivers in PHP
- PHP-MySQL Interfaces

Task 6.1:

Code:



The screenshot shows a code editor with several tabs at the top: Task6_3.php, delete.php, Task6_1.php (active), update.php, insert.php, and se. The main editor area displays the following PHP code:

```
1  <?php
2  //step 1. Database Connection + Database Selection
3  //server name, user, user's password, and database
4  //must be provided.
5  $servername = "localhost";
6  $username = "root";
7  $password = "";
8  $database = "lab6_Ex2";
9  // Create Connection
10 $conn = new mysqli($servername, $username, $password, $database);
11
12 //step 2. Check Connection
13 if ($conn->connect_error) {
14     die("Connection failed: " . $conn->connect_error);
15 }
16 echo "Connected successfully";
17 echo "<br>";
18
19 ?>
20
21 <!DOCTYPE html>
22 <html>
23 <head>
24 <title>PHP MySQLi (Object-Oriented) Basics</title>
```

```

24 <title>PHP MySQLi (Object-Oriented) Basics</title>
25 </head>
26 <body>
27 <?php
28
29 //Step 3: Perform Query
30 $query = "SELECT * FROM department";
31 $result = $conn->query($query);
32
33 //echo "dep_ID dep_Name<br>";
34 // Step 4: Display Query Results
35 if ($result) {
36     while ($row = $result->fetch_assoc()) {
37         echo $row['dep_ID'] . " " . $row['dep_Name'] . "<br>";
38     }
39     $result->free(); //Step 5:Free the result set
40 } else {
41     echo "Error executing query: " . $conn->error;
42 }
43
44 ?>
45 </body>

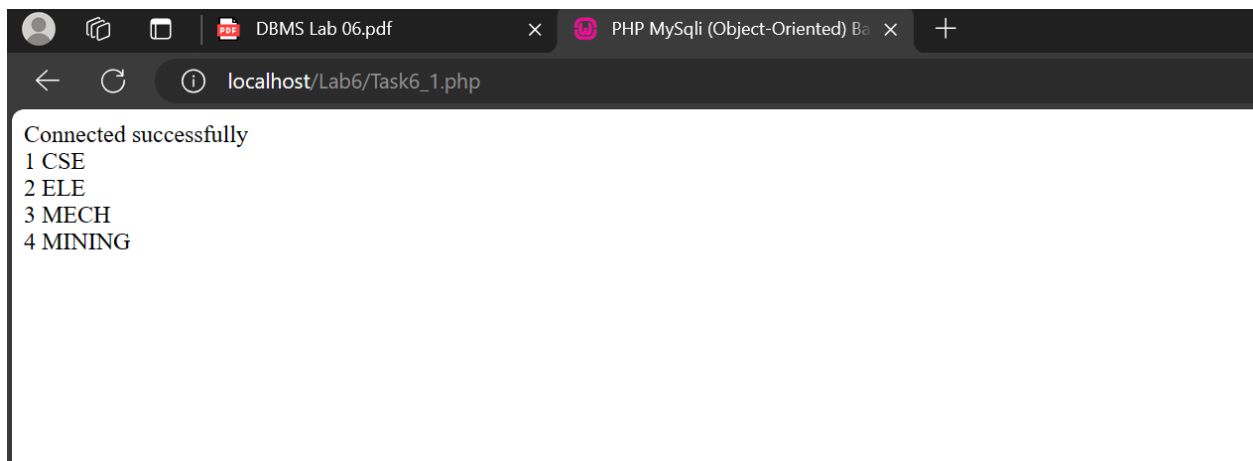
```

```

45 </body>
46 </html>
47
48 <?php
49 //step 6. Close Database Connection
50 $conn->close();
51 ?>

```

Output:



Task 6.2:

Code:

```
Task6_2.php × update.php select.php insert.php delete.php
Task6_2.php
1  <?php
2  $servername = "localhost";
3  $username = "root";
4  $password = "";
5  $dbname = "task6_2";
6
7  // Create connection
8  $conn = new mysqli($servername, $username, $password, $dbname);
9
10 // Check connection
11 if ($conn->connect_error) {
12     die("Connection failed: " . $conn->connect_error);
13 }
14 ?>
15
16 <!DOCTYPE html>
17 <html>
18 <head>
19     <title>DBMS Lab Project</title>
20 </head>
21 <body>
22
23 <h2>Insert Data into Accounts</h2>
24 <form action="insert.php" method="POST">
25     Account Number: <input type="number" name="account_number"><br>
26     Name: <input type="text" name="name"><br>
27     Balance: <input type="text" name="balance"><br>
28     <input type="submit" value="Insert">
29 </form>
30
31 <h2>Update Data in Accounts</h2>
32 <form action="update.php" method="POST">
33     Account ID to Update: <input type="text" name="account_id"><br>
34     New Name: <input type="text" name="name"><br>
35     New Balance: <input type="text" name="balance"><br>
36     <input type="submit" value="Update">
37 </form>
38
39 <h2>Delete Data from Accounts</h2>
40 <form action="delete.php" method="POST">
41     Account ID to Delete: <input type="text" name="account_id"><br>
42     <input type="submit" value="Delete">
43 </form>
44
45 <h2>Select Data from Accounts</h2>
46 <form action="select.php" method="GET">
```

```

45 <h2>Select Data from Accounts</h2>
46 <form action="select.php" method="GET">
47     Account ID to Select: <input type="text" name="account_id"><br>
48     <input type="submit" value="Select">
49 </form>
50
51 </body>
52 </html>
53

```

insert.php

```

insert.php
1 <?php
2 include 'Task6_2.php';
3
4 if ($_SERVER["REQUEST_METHOD"] == "POST") {
5     $account_number = $_POST['account_number'];
6     $name = $_POST['name'];
7     $balance = $_POST['balance'];
8
9     $sql = "INSERT INTO accounts (account_number, name, balance) VALUES ('$account_number
10
11     if ($conn->query($sql) === TRUE) {
12         echo "New record created successfully";
13     } else {
14         echo "Error: " . $sql . "<br>" . $conn->error;
15     }
16 }
17
18 $conn->close();
19 ?>
20

```

update.php

```

update.php
1 <?php
2 include 'Task6_2.php';
3
4 if ($_SERVER["REQUEST_METHOD"] == "POST") {
5     $account_id = $_POST['account_id'];
6     $name = $_POST['name'];
7     $balance = $_POST['balance'];
8
9     $sql = "UPDATE accounts SET name='$name', balance='$balance' WHERE account_number=$accou
10
11     if ($conn->query($sql) === TRUE) {
12         echo "Record updated successfully";
13     } else {
14         echo "Error: " . $sql . "<br>" . $conn->error;
15     }
16 }
17
18 $conn->close();
19 ?>
20

```

delete.php

```
delete.php
1  <?php
2  include 'Task6_2.php';
3
4  if ($_SERVER["REQUEST_METHOD"] == "POST") {
5      $account_id = $_POST['account_id'];
6
7      $sql = "DELETE FROM accounts WHERE account_number=$account_id";
8
9      if ($conn->query($sql) === TRUE) {
10         echo "Record deleted successfully";
11     } else {
12         echo "Error: " . $sql . "<br>" . $conn->error;
13     }
14 }
15
16 $conn->close();
17 ?>
```

select.php

```
select.php
1  <?php
2  include 'Task6_2.php';
3
4  if ($_SERVER["REQUEST_METHOD"] == "GET") {
5      $account_id = $_GET['account_id'];
6
7      $sql = "SELECT * FROM accounts WHERE account_number=$account_id";
8      $result = $conn->query($sql);
9
10     if ($result->num_rows > 0) {
11         echo "<h2>Account Details</h2>";
12         while($row = $result->fetch_assoc()) {
13             echo "ID: " . $row["id"]. " - Account Number: " . $row["account_number"]. " - ";
14         }
15     } else {
16         echo "0 results";
17     }
18 }
19
20 $conn->close();
21 ?>
```

Output:

```
mysql> select * from accounts;
+-----+-----+-----+
| account_number | name           | balance |
+-----+-----+-----+
| 2              | Ahmad          | 1000    |
| 5              | Ali Asghar     | 123     |
| 6              | Ali Asghar     | 123     |
| 7              | Ali Asghar Khan | 123     |
| 8              | Ali Asghar Khan | 12300   |
| 9              | Ali Asghar Khan | 12300   |
| 10             | Ali Asghar Khan | 12300   |
+-----+-----+-----+
7 rows in set (0.00 sec)
```

DBMS Lab 06.pdf

DBMS Lab Project

localhost/Lab6/Task2/insert.php

Insert Data into Accounts

Name:

Balance:

Update Data in Accounts

Account ID to Update:

New Name:

New Balance:

Delete Data from Accounts

Account ID to Delete:

Select Data from Accounts

Account ID to Select:

New record created successfully

Inserting Functionality:

Insert Data into Accounts

Name:

Balance:

10	Ali Asghar Khan	12300
11	Ahmad Ali	100000

Updating Functionality:

Update Data in Accounts

Account ID to Update:

New Name:

New Balance:

11	Ahmad Ali Khan	1000
----	----------------	------

Selecting Functionality:

Select Data from Accounts

Account ID to Select:

Account Details

ID: 10 - Account Number: 10 - Name: Ali Asghar Khan - Balance: 12300

Deleting Functionality:

```
mysql> select * from accounts;
```

account_number	name	balance
2	Ahmad	1000
5	Ali Asghar	123
6	Ali Asghar	123
7	Ali Asghar Khan	123
8	Ali Asghar Khan	12300
9	Ali Asghar Khan	12300
10	Ali Asghar Khan	12300
11	Ahmad Ali Khan	1000

Delete Data from Accounts

Account ID to Delete:

Delete

Select Data from Accounts

Account ID to Select:

Select

Record deleted successfully

```
mysql> select * from accounts;
+-----+-----+-----+
| account_number | name           | balance |
+-----+-----+-----+
| 2              | Ahmad         | 1000    |
| 5              | Ali Asghar    | 123     |
| 6              | Ali Asghar    | 123     |
| 7              | Ali Asghar Khan | 123     |
| 8              | Ali Asghar Khan | 12300   |
| 9              | Ali Asghar Khan | 12300   |
| 11             | Ahmad Ali Khan | 1000    |
+-----+-----+-----+
7 rows in set (0.00 sec)
```

Task 6.3:

Code:

```
Task6_3.php X delete.php Task6_1.php update.php insert.php select.php
Task6_3.php
1  <?php
2
3  $servername = "localhost";
4  $username = "root";
5  $password = "";
6  $dbname = "task6_2";
7
8  try {
9      $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
10     // Set the PDO error mode to exception
11     $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
12 } catch (PDOException $e) {
13     echo "Connection failed: " . $e->getMessage();
14 }
15 ?>
16
17 <!DOCTYPE html>
18 <html>
19 <head>
20     <title>DBMS Lab Project</title>
21 </head>
22 <body>
23
24 <h2>Insert Data into Accounts</h2>
```

```

25 <form action="insert.php" method="POST">
26     Name: <input type="text" name="name"><br>
27     Balance: <input type="text" name="balance"><br>
28     <input type="submit" value="Insert">
29 </form>
30
31 <h2>Update Data in Accounts</h2>
32 <form action="update.php" method="POST">
33     Account ID to Update: <input type="text" name="account_id"><br>
34     New Name: <input type="text" name="name"><br>
35     New Balance: <input type="text" name="balance"><br>
36     <input type="submit" value="Update">
37 </form>
38
39 <h2>Delete Data from Accounts</h2>
40 <form action="delete.php" method="POST">
41     Account ID to Delete: <input type="text" name="account_id"><br>
42     <input type="submit" value="Delete">
43 </form>
44
45 <h2>Select Data from Accounts</h2>
46 <form action="select.php" method="GET">
47     Account ID to Select: <input type="text" name="account_id"><br>
48     <input type="submit" value="Get Data">

```

insert.php

```

insert.php
1  <?php
2  include 'Task6_3.php';
3
4  if ($_SERVER["REQUEST_METHOD"] == "POST") {
5      $account_number = $_POST['account_number'];
6      $name = $_POST['name'];
7      $balance = $_POST['balance'];
8
9      try {
10         $sql = "INSERT INTO accounts (account_number, name, balance) VALUES (:account_number, :name, :balance)";
11         $stmt = $conn->prepare($sql);
12         $stmt->bindParam(':account_number', $account_number);
13         $stmt->bindParam(':name', $name);
14         $stmt->bindParam(':balance', $balance);
15         $stmt->execute();
16         echo "New record created successfully";
17     } catch (PDOException $e) {
18         echo "Error: " . $e->getMessage();
19     }
20 }
21
22 $conn = null;
23 ?>

```

update.php

```
Task6_3.php delete.php Task6_1.php update.php X insert.php select.php
update.php
1  <?php
2  include 'Task6_3.php';
3
4  if ($_SERVER["REQUEST_METHOD"] == "POST") {
5      $account_id = $_POST['account_id'];
6      $name = $_POST['name'];
7      $balance = $_POST['balance'];
8
9      try {
10         $sql = "UPDATE accounts SET name=:name, balance=:balance WHERE account_number=:acco
11         $stmt = $conn->prepare($sql);
12         $stmt->bindParam(':account_id', $account_id);
13         $stmt->bindParam(':name', $name);
14         $stmt->bindParam(':balance', $balance);
15         $stmt->execute();
16         echo "Record updated successfully";
17     } catch (PDOException $e) {
18         echo "Error: " . $e->getMessage();
19     }
20 }
21
22 $conn = null;
23 ?>
```

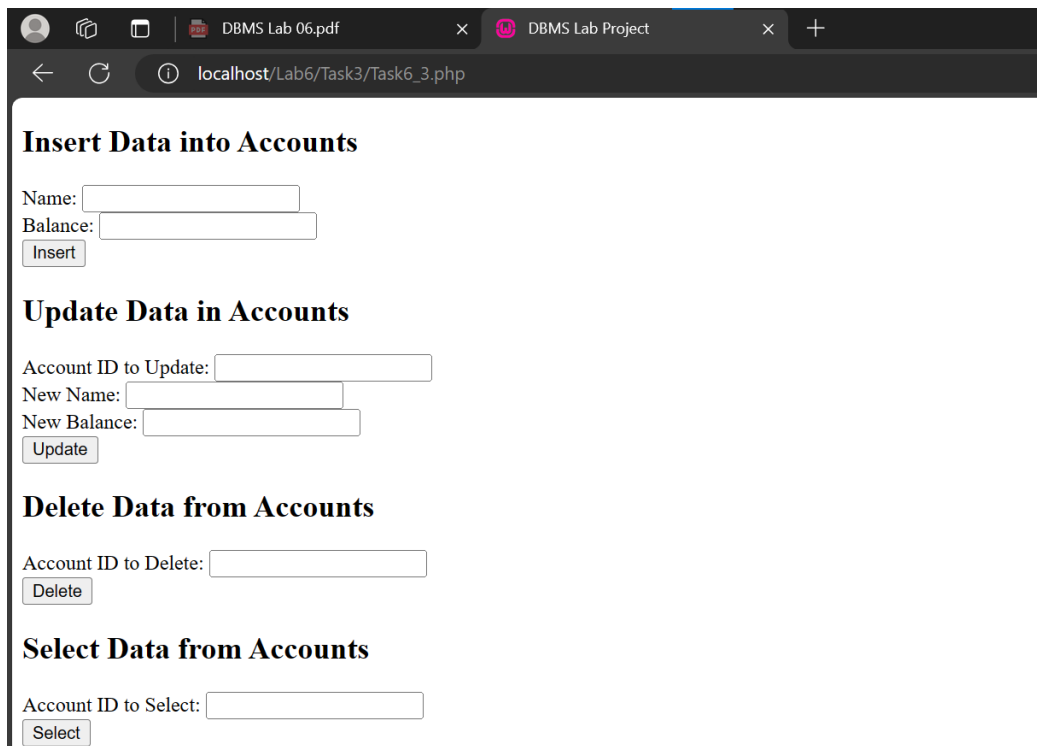
delete.php

```
Task6_3.php delete.php X Task6_1.php update.php insert.php select
delete.php
1  <?php
2  include 'Task6_3.php';
3
4  if ($_SERVER["REQUEST_METHOD"] == "POST") {
5      $account_id = $_POST['account_id'];
6
7      try {
8         $sql = "DELETE FROM accounts WHERE account_number=:account_id";
9         $stmt = $conn->prepare($sql);
10         $stmt->bindParam(':account_id', $account_id);
11         $stmt->execute();
12         echo "Record deleted successfully";
13     } catch (PDOException $e) {
14         echo "Error: " . $e->getMessage();
15     }
16 }
17
18 $conn = null;
19 ?>
```

select.php

```
select.php
1 <?php
2 include 'Task6_3.php';
3 if ($_SERVER["REQUEST_METHOD"] == "GET") {
4     $account_id = $_GET['account_id'];
5     try {
6         $sql = "SELECT * FROM accounts WHERE account_number=:account_id";
7         $stmt = $conn->prepare($sql);
8         $stmt->bindParam(':account_id', $account_id);
9         $stmt->execute();
10        $result = $stmt->fetchAll();
11
12        if (count($result) > 0) {
13            echo "<h2>Account Details</h2>";
14            foreach ($result as $row) {
15                echo "ID: " . $row["account_number"]. " - Account Number: " . $row["account_
16            }
17        } else {
18            echo "0 results";
19        }
20    } catch (PDOException $e) {
21        echo "Error: " . $e->getMessage();
22    }
23 }
24 $conn = null;
```

Output:



The screenshot shows a web browser with two tabs: "DBMS Lab 06.pdf" and "DBMS Lab Project". The address bar shows "localhost/Lab6/Task3/Task6_3.php". The page content is as follows:

Insert Data into Accounts

Name:
Balance:

Update Data in Accounts

Account ID to Update:
New Name:
New Balance:

Delete Data from Accounts

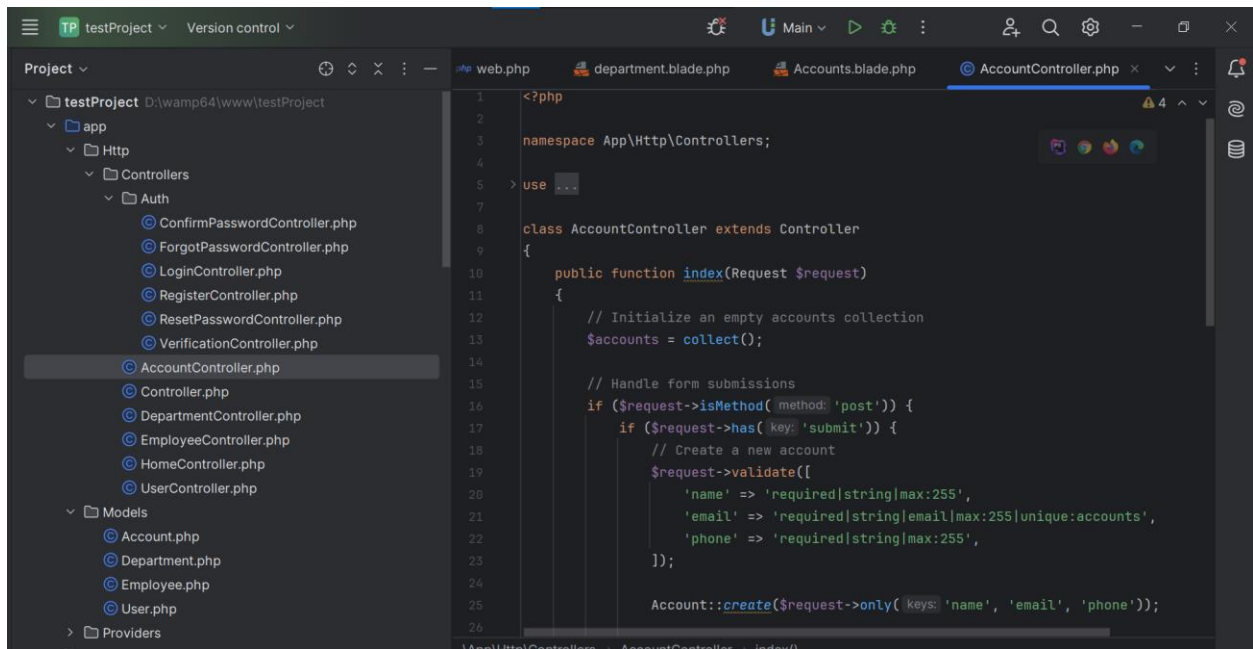
Account ID to Delete:

Select Data from Accounts

Account ID to Select:

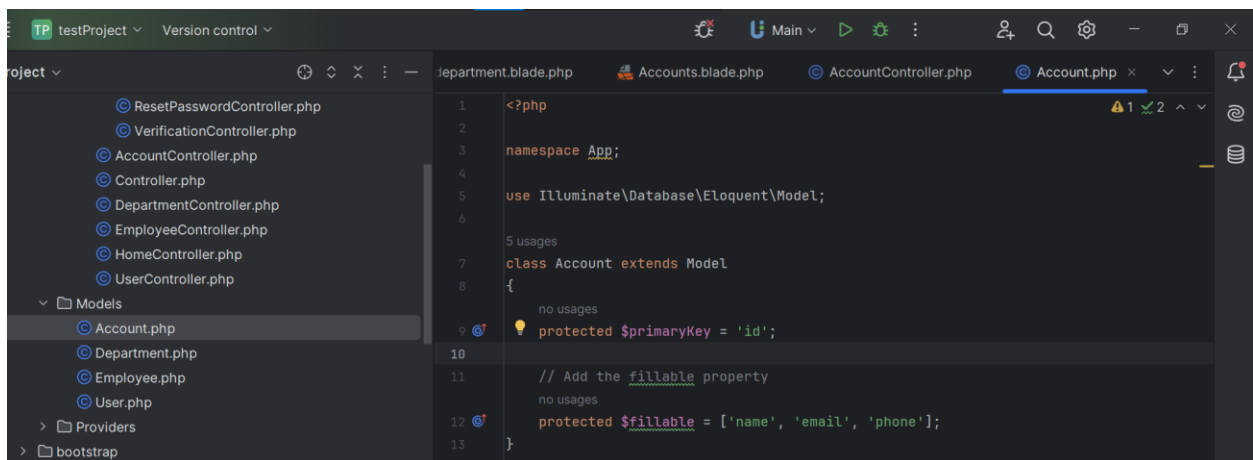
Task 6.4:

Code:



The screenshot shows the Visual Studio Code editor with the 'testProject' workspace. The file explorer on the left shows the project structure: 'app' > 'Http' > 'Controllers' > 'Auth'. The 'AccountController.php' file is selected. The code in the editor is as follows:

```
1 <?php
2
3 namespace App\Http\Controllers;
4
5 use ...
6
7
8 class AccountController extends Controller
9 {
10     public function index(Request $request)
11     {
12         // Initialize an empty accounts collection
13         $accounts = collect();
14
15         // Handle form submissions
16         if ($request->isMethod('post')) {
17             if ($request->has('submit')) {
18                 // Create a new account
19                 $request->validate([
20                     'name' => 'required|string|max:255',
21                     'email' => 'required|string|email|max:255|unique:accounts',
22                     'phone' => 'required|string|max:255',
23                 ]);
24
25                 Account::create($request->only('name', 'email', 'phone'));
26             }
27         }
28     }
29 }
```



The screenshot shows the Visual Studio Code editor with the 'testProject' workspace. The file explorer on the left shows the project structure: 'Models'. The 'Account.php' file is selected. The code in the editor is as follows:

```
1 <?php
2
3 namespace App;
4
5 use Illuminate\Database\Eloquent\Model;
6
7
8 class Account extends Model
9 {
10     // Add the fillable property
11     protected $fillable = ['name', 'email', 'phone'];
12 }
13
```

```
<?php
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

class CreateAccountsTable extends Migration
{
    public function up()
    {
        Schema::create( table: 'accounts', function (Blueprint $table) {
            $table->increments( column: 'id');
            $table->string( column: 'name');
            $table->string( column: 'email')->unique();
            $table->string( column: 'phone');
            $table->timestamps();
        });
    }

    public function down()
    {
        Schema::dropIfExists( table: 'accounts');
    }
}
```

```
<!DOCTYPE html>
<html>
<head>
<title>Accounts</title>
</head>
<body>
@if(session('success'))
<div>{{ session('success') }}</div>
@endif

<h1>Accounts</h1>

<h2>Add New Account</h2>
<form method="post" action="{{ route('accounts') }}">
    @csrf
    Name: <input type="text" name="name"><br>
    Email: <input type="email" name="email"><br>
    Phone: <input type="text" name="phone"><br>
    <input type="submit" name="submit" value="Add Account">
</form>

<h2>Update Account</h2>
<form method="post" action="{{ route('accounts') }}">
    @csrf
    ID: <input type="text" name="id"><br>
```

```
Route::get( uri: '/accounts', [AccountController::class, 'index']->name( name: 'accounts'
```

Output:



Accounts

Add New Account

Name:

Email:

Phone:

Update Account

ID:

Name:

Email:

Phone:

Delete Account

ID:

Show All Entries